

5 Control of heartbeat is myogenic. This means the electrical activity controlling the rhythm of a regular heartbeat begins in the heart muscle itself.

Atrial fibrillation (AF) is an abnormal heart rhythm that causes rapid and irregular contractions of the atria. Untreated cases of AF can lead to a stroke.

- (a)** A stroke is caused when a small blood clot, often forming in the left atrium, is carried by the blood to the brain where it blocks a small artery and leads to brain damage.
 - (i)** List all of the structures through which a blood clot in the left atrium must travel to reach the blood vessels supplying the brain.

The structures must be listed in the correct sequence.

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..... [1]

- (ii)** Explain why blocking a small artery in the brain leads to brain damage.

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..... [1]

- (b)** A common cause of AF is when a small group of muscle cells in the wall of the left atrium starts to send out electrical impulses to the surrounding heart muscle cells.

Explain how the control of heartbeat by the sinoatrial node can be disrupted by AF, resulting in rapid and irregular atrial contractions.

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- (c) Red blood cells are involved in the transport of oxygen and carbon dioxide in the blood.

Fig. 5.1 is a diagram representing the exchange of oxygen and carbon dioxide between a red blood cell in a capillary and a respiring cell. Some of the reactions that take place in the red blood cell are also shown. The diagram is **not** drawn to scale.

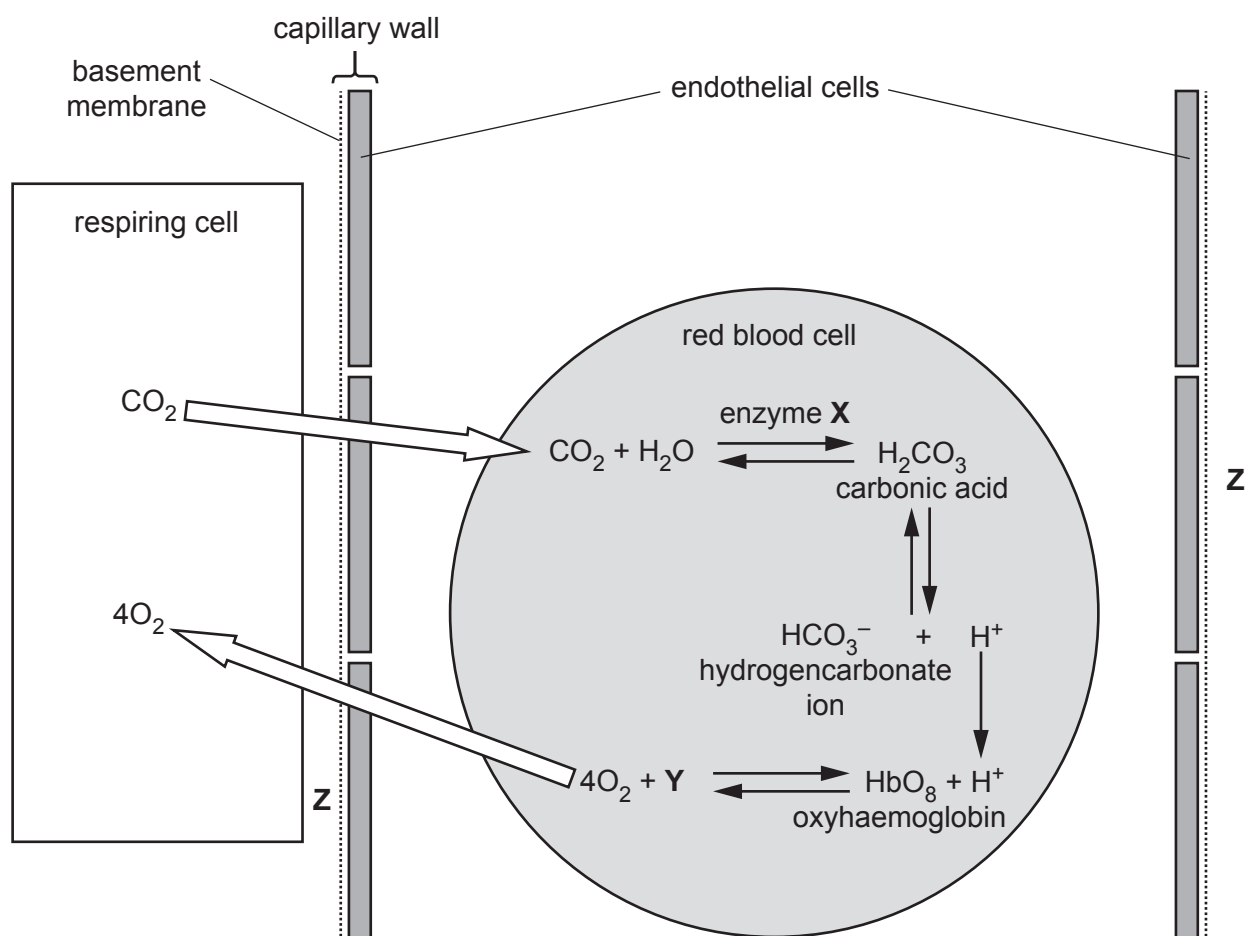


Fig. 5.1

- (i) Identify enzyme **X** and molecule **Y** in Fig. 5.1.

X

Y

[2]

- (ii) The hydrogencarbonate ions shown in Fig. 5.1 leave the red blood cell and are replaced by chloride ions.

State why it is necessary for chloride ions to enter the red blood cell as hydrogencarbonate ions leave.

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(d) Identify the aqueous environment, labelled **Z** in Fig. 5.1, that surrounds the respiring cell.

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(e) Oxygen and carbon dioxide are also exchanged between blood capillaries and alveoli in the lungs.

The gas exchange system has specialised cells to prevent harmful microscopic particles that are present in inhaled air from reaching the alveoli.

These particles are associated with many respiratory diseases.

Explain how specialised cells in the gas exchange system prevent harmful microscopic particles from reaching the alveoli.

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..... [3]

[Total: 12]